
Approved ANSI Standard

ANSI/VITA 51.1 – 2013 (R2018)

Reliability Prediction MIL-HDBK-217 Subsidiary Specification

Abstract

This specification provides standard defaults and methods to adjust the models in MIL-HDBK-217F Notice 2. This is not a revision of MIL-HDBK-217F Notice 2 but a standardization of the inputs to the MIL-HDBK-217F Notice 2 calculations to give more consistent results.



Approved: October 16, 2013

Reaffirmed: October 10, 2018

American National Standards Institute, Inc.

Secretariat: VITA

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Printed in the United States of America - R1.3, ISBN 1-885731-47-7.

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Foreword

When planning military or high reliability applications, knowledge is needed regarding the potential life of electronic components involved in that system. More and more of the electronic components used in such systems are Commercial Off-The-Shelf (COTS) items. As such, the military program has very little control of the design, development, and availability of these modules. A reliable method to predict the failure rate of these systems is essential for planning sparing.

Reliability is defined as the ability of a product to perform its intended function under stated conditions for a specific period of time. Reliability is considered to be an important aspect of any electronic system, especially when utilizing COTS products. Reliability assurance is a very critical element in the acquisition and life-cycle management of military systems and especially challenging with the broad use of COTS items.

In 1961, the first edition of MIL-HDBK-217 [1] was published, providing a basic failure rate prediction methodology that is still in widespread use today, despite problems, limitations and inaccuracies that have been identified with the MIL-HDBK-217F Notice 2 methodology.

Manufacturers and electronic reliability engineers use different methods to adjust the models in MIL-HDBK-217F Notice 2 for newer technologies, use different defaults for unknown stress conditions, and make differing assumptions of quality and complexity factors for COTS items. These differing methods yield results that are not comparable. This specification is intended to provide a standard method for reliability engineers to perform failure rate predictions for COTS items used in military or high reliability applications.

1 Introduction

1.1 Objectives

Reliability engineers in government and industry continue to use MIL-HDBK-217F Notice 2 to perform failure rate predictions. Because it has not been updated recently, there are many new developments in electronics that are not accounted for, so reliability engineers must make changes to the MIL-HDBK-217F Notice 2 methodology to make their failure rate predictions more realistic.

This specification provides a standard method of performing reliability predictions on COTS modules using MIL-HDBK-217F Notice 2. This is not a revision or a modification of MIL-HDBK-217F Notice 2 but a standardization of the inputs to the MIL-HDBK-217F Notice 2 calculations to give more consistent MTBF numbers. This specification will assist the user in determining the credibility of published failure rate information and outline the techniques necessary for developing and applying this information to military or high reliability applications.

It isn't possible to predict the reliability of counterfeit parts. Therefore, use of this specification is only valid if there are measures taken to assure the electronics are free of counterfeit parts. See Appendix B for more information on counterfeit parts.

1.2 Overview

This specification provides standard defaults and methods to adjust the models in MIL-HDBK-217F Notice 2. This is not a revision of MIL-HDBK-217F Notice 2 but a standardization of the inputs to the MIL-HDBK-217F Notice 2 calculations to give more consistent results. It documents the rationale for each default or adjustment factor provided, and requires the documentation of any further modifications to the models that are used.

It is not the intent for this specification to be used to judge the accuracy or require the choice of this or any other failure rate prediction method. That decision is driven by contractual or market issues. If a MIL-HDBK-217F Notice 2 prediction is required or chosen for some reason, this specification provides standard defaults and methods to adjust the models in MIL-HDBK-217F Notice 2. It also requires the documentation of deviations from these standards, so that credibility of the results can be maintained.

If this specification doesn't specify a new factor for a particular part type, then the published MIL-HDBK-217F Notice 2 is to be used for that type of part because alternative defaults have not been identified or substantiated.

1.3 Terminology

1.3.1 Specification Key Words

To avoid confusion and to make very clear what the requirements for compliance are, many of the paragraphs in this standard are labeled with keywords that indicate the type of information they contain. These keywords are listed below:

- Rule
- Recommendation
- Suggestion
- Permission
- Observation

Any text not labeled with one of these keywords should be interpreted as descriptive in nature. These will be written in either a descriptive or a narrative style.

The keywords are used as follows:

Rule <chapter>-<number>:

Rules form the basic framework of this standard. They are sometimes expressed in text form and sometimes in the form of figures, tables or drawings. All rules shall be followed to ensure compatibility between board and backplane designs. All rules use the “shall” or “shall not” words to emphasize the importance of the rule. The “shall” or “shall not” words are reserved exclusively for stating rules in this standard and are not used for any other purpose.

Recommendation <chapter>-<number>:

Wherever a recommendation appears, designers would be wise to take the advice given. Doing otherwise might result in poor performance or awkward problems. Recommendations found in this standard are based on experience and are provided to designers to speed their traversal of the learning curve. All recommendations use the “should” or “should not” words to emphasize the importance of the recommendation. The “should” or “should not” words are reserved exclusively for stating recommendations in this standard and are not used for any other purpose.

Suggestion <chapter>-<number>:

A suggestion contains advice, which is helpful but not vital. The reader is encouraged to consider the advice before discarding it. Some design decisions that need to be made are difficult until experience has been gained. Suggestions are included to help a designer who has not yet gained this experience.

Permission <chapter>-<number>:

In some cases a rule does not specifically prohibit a certain design approach, but the reader might be left wondering whether that approach might violate the spirit of the rule or whether it might lead to some subtle problem. Permissions reassure the reader that a certain approach is acceptable and will cause no problems. All permissions use the “may” words to emphasize the importance of the permission. The lower-case “may” words are reserved exclusively for stating permissions in this standard and are not used for any other purpose.

Observation <chapter>-<number>:

Observations do not offer any specific advice. They usually follow naturally from what has just been discussed. They spell out the implications of certain rules and bring attention to things that might otherwise be overlooked. They also give the rationale behind certain rules so that the reader understands why the rule must be followed.

1.3.2 Definitions

The following terms are used within the body of the specification. In this context, they have the following meanings.

Term	Definition
FIT or FITs	“Failures-In-Time” a unit of failure rate, Failures per 10 ⁹ hours, Failures per billion hours (Note: MIL-HDBK-217F Notice 2 doesn’t use “FITs” but instead uses failure per million hours (FPMH))
FPMH	Failures per million hours
Known Pedigree	Components are considered to be of “known pedigree” if they are procured, verified and managed using recognized standard processes that have been approved by the prime contractor or certified by an industry recognized third party certification agency. (see Appendix C for guidance)
MTBF	Mean Time Between Failures – a population statistic: For a particular interval, the total functioning life of a population of an item divided by the total number of failures within the population during the measurement

	interval. The definition holds for time, cycles, miles, events, or other measures of life units. (MIL-STD-721)
QML	Qualified Manufacturers List
QPL	Qualified Parts List
RAC	Reliability Analysis Center
SRC	System Reliability Center
Time-to-failure	The expected life of a unit. May also be expressed as Cycles-to-failure or as expected life.
Wearout failure	Failure caused by fatigue or aging of a unit.

Table 1.3.2-1: Term Definitions

1.3.3 Quality of Data Sources

Based on the experience of industry reliability engineers and reliability prediction specialists who collaborated in the development of this specification, default settings and detailed guidelines to use when developing adjustment factors are defined in this specification. These default settings and adjustment guidelines can be found in section 2 of this document.

Data sources are cited where the specified default value is based on field data or test data analysis. Source data that was used to derive the specified factor is provided in the VITA 51 website, and links provided in the electronic version of this document. In some cases, the default value is based on engineering judgment, and this is noted where applicable. To be conservative, PiQs based on engineering judgment were generally set to higher values than the PiQs that were derived from field or test data.

Because the specified adjustment factors come from a variety of sources, the following categories are used throughout this document to make clear to the user what quality of substantiation is behind each specified factor.

Category 1. Engineering Judgment

Category 2. Based on Field Experience Data

Category 3. Based on Test Data

Users of this specification are invited to provide new information for future revisions, by registering with the VITA 51 Community of Practice (CoP), as described in the main specification VITA 51.0, and by sharing experience and data that will improve the methods provided in this specification.

1.4 Normative References

- [VITA 51.0] ANSI/VITA 51.0-2008 (R2012), "American National Standard for Reliability Prediction" VITA, 2012
- [VITA 51.2] ANSI/VITA 51.2-2011, "Physics of Failure Reliability Predictions" VITA, 2011
- [VITA 51.3] ANSI/VITA 51.3-2010, "Qualification and Environmental Stress Screening in Support of Reliability Predictions" VITA, 2010

1.5 Non-Normative References

- [1] Military Handbook (MIL-HDBK)-217F Notice 2, Reliability Prediction of Electronic Equipment, February 28, 1995
- [2] "Microelectronic Reliability: Physics of Failure Based Modeling and Lifetime Evaluation, Mark White and Joseph B. Bernstein, JPL Publication 08-5 2/08, 2008 February
- [3] "Reliability Evaluation of Plastic Encapsulated Parts" IEEE Transactions on Reliability, Vol. 42 No. 4, Leonard Weil, Michael Pecht, Ed Hakim, 1993 December
- [4] "Failure rate for various components", See VITA website, <https://www.vita.com/VITA-51-Family>
- [5] "Field data for resistor failures", See VITA website, <https://www.vita.com/VITA-51-Family>
- [6] Chip Caps and Chip Caps Roll up, See VITA website, <https://www.vita.com/VITA-51-Family>
- [7] Blatta, N., Gormally, P., Iannaccone, V., Harvilchuck, L, and Hillman, C. "Robustness of Surface Mount Multilayer Ceramic Capacitors Assembled with Pb-Free Solder" http://www.dfrsolutions.com/wp-content/uploads/2012/06/2005_Capcracking_Whitepaper.pdf
- [8] Reliability Toolkit: Commercial Practices Edition. RAC, Alion, 1995

2 Standard Methods for MIL-HDBK-217F Notice 2

This specification provides standard methods for using MIL-HDBK-217F Notice 2 for COTS electronics modules in military or high reliability applications, including a set of standard defaults and standard methods for developing adjustment factors.

A user may have good reason to use different defaults or different adjustment factors for their reliability predictions. If that happens, a disclosure statement (VITA 51.0 Appendix A) is required by VITA 51.0 Rule 2.5-1 to document these differences.

Default values for pi-factors are only used in the absence of better information.

Recommendation 2-1: Once detailed design, quality, usage and environmental information are available, defaults SHOULD NOT be used. Pi-factors reflecting application environments used as inputs to the MIL-HDBK-217F Notice 2 reliability predictions SHOULD be documented and reported as required by the program contract.

Recommendation 2-2: Standard defaults are provided for both Parts Stress and Parts Count methods of failure rate prediction. The appropriate method SHOULD be selected based on the level of detail design information available to the user, and the method chosen SHOULD be indicated on the disclosure statement (ref. VITA 51.0 Rule 2.5-1) accompanying a prediction report.

2.1 Parts Stress Method Standard Defaults

For each standard default provided in this section, the categories defined in 1.3.3 are noted to clarify the quality of derivation. Source data for the derived factors are provided on the VITA website in the Reliability Marketing Alliance, Frequently Asked Questions (FAQ) area, and referenced where applicable. Links to the data are provided in the electronic copy of this document. If the embedded links in this document don't work, contact VITA.

2.1.1 General

Rule 2.1.1-1: The reliability prediction SHALL be associated with a stated temperature and environment.

Observation 2.1.1-1: Most COTS item predictions are expressed with 30 degrees C and Ground Benign environments.

2.1.2 Integrated Circuits (IC)

Observation 2.1.2-1: Random failure rate models for Integrated Circuits (IC) can be found in section 5 of MIL-HDBK-217F Notice 2.

Observation 2.1.2-2: As integrated circuit technologies advance, functional densities increase and feature sizes decrease. This has been observed to cause a higher susceptibility to wearout effects and single event effects. [2] For technologies with less than 130nm feature sizes, wearout effects may occur much earlier than for older technologies.

Recommendation 2.1.2-1: For technologies with less than 130nm feature sizes, the wearout models that are provided in ANSI/VITA 51.2-2011 or equivalent SHOULD be used. The wearout mechanisms to be modeled include electromigration, time-dependent dielectric breakdown, hot carrier injection and negative bias temperature instability.

Observation 2.1.2-3: The MIL-HDBK-217F Notice 2 "Quality Level" for commercial integrated circuits has a PiQ of 10. This PiQ of 10 is not representative of the quality of most commercial integrated circuit product lines.

Rule 2.1.2-1: For commercial integrated circuits of known pedigree, PiQ Value = 1 SHALL be used. (Category 3 – see Reference [3])

Observation 2.1.2-4: In MIL-HDBK-217F Notice 2, the maximum number of gates (60,000), maximum number of transistors (10,000) is several orders of magnitude too low for most current technology ICs,

and future technology will far exceed what is available today. There are several common approaches to handling greater gates and transistors. MIL-HDBK-217F Notice 2 has a note in section 5.1 "For CMOS gate counts above 60,000 use the VHSIC/VHSIC-Like model in Section 5.3." This doesn't cover what to do for Bipolar, Linear ICs or memory devices. One commonly used approach is to use the maximum C1 value for anything greater. Another common approach is to use the complexity equation that was used to create the C1 tables, to mathematically extrapolate the values for however many gates, transistors or pins the part has. These different approaches will cause differences in the results, and confusion if not disclosed and clarified. Although not explicitly provided in MIL-HDBK-217F Notice 2, the extrapolation approach is permitted (see Permission 2.1.2-1).

Rule 2.1.2-2: For non-CMOS parts with more gates than 60,000, and for Linear parts with more than 10,000 transistors in MIL-HDBK-217F Notice 2 Section 5 the method used for determining the complexity factors for the reliability prediction SHALL be stated in the reliability disclosure form (VITA 51.0, Appendix A).

Permission 2.1.2-1: For non-CMOS parts with more gates than 60,000, and for Linear parts with more than 10,000 transistors, the complexity factors for C1 MAY be based on the extrapolated values provided in Table 2.1.2-1. (Category 1)

Table 2.1.2-1. Additional Complexity Factors (C1) for Microcircuits, for complexities exceeding those provided in MIL-HDBK-217F Notice 2 Section 5.1

Bipolar Digital and Linear Gate/Logic Array Die C1 Factors						
Linear		PLA/PAL				
No. Transistors	C1	No. Gates		C1	Source:	
10,001 to 30,000	0.08	5,001 to 10,000		0.084	Category 1	
30,001 to 60,000	0.10	10,001 to 30,000		0.168	Category 1	
		30,001 to 60,000		0.336	Category 1	
MOS Digital and Linear Gate/Logic Array Die C1 Factors						
Linear		PLA/PAL				
No. Transistors	C1	No. Gates		C1	Source:	
10,001 to 30,000	0.080	20,001 to 30,000		0.0136	Category 1	
30,001 to 60,000	0.100	30,001 to 60,000		0.0272	Category 1	
Microprocessor						
No. Bits	Bipolar C1	MOS C1				
Up to 64	.48	1.12				Category 1
Up to 128	.96	2.24				Category 1
Memories						
Memory Size, B (Bits)	MOS				Bipolar	
	ROM	PROM, UVEPROM, EEPROM, EAPROM	DRAM	SRAM (MOS & BiMOS)	ROM, PROM	SRAM
1 M < B ≤ 4 M	0.0104	0.0136	0.020	0.124	0.150	0.084
4 M < B ≤ 16 M	0.0208	0.0272	0.040	0.248	0.300	0.168

16 M < B ≤ 64 M	0.0416	0.0544	0.080	0.496	0.600	0.336
64 M < B ≤ 256 M	0.0832	0.1088	0.160	0.992	1.20	0.672

Observation 2.1.2-5: Some commonly used reliability tools truncate C2 at the maximum numbers provided in the table in section 5.9. The MIL-HDBK-217F Notice 2 section however, does provide equations that have no stated limits.

Suggestion 2.1.2-1: For parts with more pins than the values provided in MIL-HDBK-217F Notice 2 section 5.9 table for C2, use the equations provided below that table to derive further values for C2 for parts with higher pin counts.

Observation 2.1.2-6: The MIL-HDBK-217F Notice 2 approach to modeling IC failure rates uses factors such as “number of gates” or “number of transistors” that don’t appear to drive empirical reliability data on current technology ICs as published by semiconductor manufacturers. Published failure rates tend to be based more on fabrication or process technology, and published as “FIT rates” (“Failures In Time” or failures per 10⁹ hours).

Permission 2.1.2-2: Manufacturers’ published failure rates MAY be used for IC failure rates instead of a prediction based on MIL-HDBK-217F Notice 2. This will be included in the required disclosure statement, in accordance with VITA 51.0 Rule 2.5-1.

Rule 2.1.2-3: “Package Type” = Nonhermetic: DIPs, PGA, SMT (most commercial ICs are Plastic Encapsulated Microcircuits (PEMs) = Nonhermetic) SHALL be used for commercial integrated circuits.

Suggestion 2.1.2-2: Enter the “Years in Production” as this affects the failure rate calculation.

Rule 2.1.2-4: Whenever a temperature rise calculation (e.g. case to junction) is used, the reliability prediction report SHALL include information indicating the source of the derivation.

Recommendation 2.1.2-2: If operating power and/or thermal resistance data are not available the maximum power dissipation from the specification for the closest equivalent device SHOULD be used (as per MIL-HDBK-217F Notice 2 page 5-17). (Category 1)

Rule 2.1.2-5: “Type” = DRAM SHALL be used for Memory SDRAM devices.

Rule 2.1.2-6: “Type” = SRAM SHALL be used for Memory NVSRAM devices.

Rule 2.1.2-7: “Type” = EEPROM, Flotox SHALL be used as default for Memory Flash devices.

2.1.3 Semiconductor

Observation 2.1.3-1: Semiconductors can be found in section 6 of MIL-HDBK-217F Notice 2. Section 6, semiconductors are used in the case of discrete components like discrete transistors, diodes, etc.

Rule 2.1.3-1: Quality Level (PiQ = 1) SHALL be used as a standard default for commercial semiconductors of known pedigree. (Category 1)

Rule 2.1.3-2: When the actual voltage ratio is unknown, “Voltage Ratio” = 0.5 (50 percent) SHALL be used as a standard default for semiconductors, if the model has a PiS, voltage stress factor. This is considered an average setting for the voltage ratio. (Category 1)

Recommendation 2.1.3-1: Based on the analysis provided in Appendix D, a base failure rate, λ_b = 0.0012 SHOULD be used for MOSFETs of low frequency (similar to MIL-PRF-19500). (Category 3, based on 60% confidence)

Recommendation 2.1.3-2: Based on the analysis provided in Appendix D, a base failure rate, λ_b = 0.006 SHOULD be used for MOSFETs of high frequency (similar to MIL-PRF-19500). (Category 3, based on 60% confidence)

2.1.4 Resistor

Observation 2.1.4-1: Resistors can be found in section 9 of MIL-HDBK-217F Notice 2.

Observation 2.1.4-2: The “Quality Level” for a Commercial resistor has a PiQ of 10. This PiQ of 10 is not representative of the quality of most commercial resistor product lines.

Rule 2.1.4-1: “Quality Level” of PiQ = 1 SHALL be used as a standard default for commercial resistors of known pedigree (Category 1), except when using the values provided in Rule 2.1.4-2.

Rule 2.1.4-2: “Quality Level” of PiQ = 0.1 SHALL be used as a standard default for commercial resistors of known pedigree similar to type RM resistors of MIL-PRF-55342 (Resistor, Fixed, Film, Chip) and similar to type RZ resistors of MIL-PRF-83401 (Resistor Networks, Fixed, Film). (RM: Category 2, reference [4]) (RZ: Category 3, reference [5])

Rule 2.1.4-3: “Power Ratio” = 0.5 (50 percent) SHALL be used as a standard default for resistors unless a power ratio is known. (Category 1) This is considered an average setting for the voltage ratio. Ensure that the power rating and type of the resistor is correct.

2.1.5 Capacitor

Observation 2.1.5-1: Capacitors can be found in section 10 of MIL-HDBK-217F Notice 2.

Observation 2.1.5-2: The “Quality Level” for a Commercial capacitor has a PiQ of 10. This PiQ of 10 is not representative of the quality of most commercial capacitor product lines.

Rule 2.1.5-1: “Quality Level” PiQ = 1 SHALL be used as a standard default for commercial capacitors of known pedigree, except when using the values provided in Rules 2.1.5-2 to 2.1.5-8. (Category 1)

Rule 2.1.5-2: “Quality Level” PiQ = 0.1 SHALL be used as a standard default for commercial capacitors of known pedigree similar to type CDR capacitors of MIL-PRF-55681 (Capacitor, Chip, Fixed, Ceramic). (Category 2, reference [6])

Rule 2.1.5-3: “Quality Level” PiQ = 0.1 SHALL be used as a standard default for commercial capacitors of known pedigree, similar to type PS capacitors of MIL-PRF-49470 (Capacitors, Fixed, Ceramic, Switch Mode Power Supply, Horizontally Stacked) (Category 1, based on test data of individual ceramic chip capacitors, see Rule 2.1.5-2)

Rule 2.1.5-4: “Quality Level” PiQ = 0.1 SHALL be used as a standard default for commercial capacitors similar to type CWR capacitors of MIL-PRF-55365 (Capacitor, Fixed, Electrolytic Tantalum, Chip) of known pedigree, but only for those with capacitance $C \geq 0.1 \mu\text{F}$. (Category 1, see Appendix E)

Rule 2.1.5-5: “Quality Level” PiQ = 0.1 SHALL be used as a standard default for commercial capacitors of known pedigree, similar to type CKR capacitors of MIL-PRF-39014 (Capacitor, Fixed, Ceramic, general purpose). (Category 2: reference [4])

Rule 2.1.5-6: “Quality Level” PiQ = 0.1 SHALL be used as a standard default for commercial capacitors of known pedigree, similar to type CSR capacitors of MIL-PRF-39003 (Capacitor, Fixed, Solid Electrolyte, Tantalum). (Category 2: reference [4])

Rule 2.1.5-7: “Quality Level” PiQ = 0.1 SHALL be used as a standard default for commercial capacitors of known pedigree, similar to type CLR capacitors of MIL-PRF-39006 (Capacitor, Fixed, Non- Solid Electrolyte, Tantalum). (Category 1)

Rule 2.1.5-8: “Quality Level” PiQ = 0.46 SHALL be used as a standard default for commercial capacitors of known pedigree, similar to type CCR capacitors of MIL-PRF-20 (Capacitor, Fixed, Ceramic, Temperature Compensating) (Category 2, reference [4])

Rule 2.1.5-9: When the voltage ratio is unknown, “Voltage Ratio” = 0.5 (50 percent) SHALL be used as a standard default for capacitors. (Category 1) This is considered an average setting for the voltage ratio. Ensure the capacitance value and capacitance units are correct.

Permission 2.1.5-1: In the case of functionally parallel capacitors, e.g. bypass capacitors, other considerations MAY be made such as a reliability credit for open circuit or for change-of-value failures, but these must be disclosed as required by VITA 51.0 Rule 2.5-1.

Observation 2.1.5-3: The vast majority of failures of multilayer ceramic chip capacitors are from cracks that result in a short.[7] More than half of these failures are not inherent to the part, but related to manufacturing and assembly processes and to the physical stability of the circuit board during in-circuit test and customer use. Failures inherent to the part may vary dramatically depending on the part manufacturer and special features of the part design. The quality factor upgrade for prediction purposes may not be meaningful unless we consider all of these factors.

Observation 2.1.5-4: Because ceramic capacitors can represent the largest portion of the predicted circuit board failure rate, part selection and an understanding of the failure mechanisms is especially important.

2.1.6 Inductor

Observation 2.1.6-1: Inductors can be found in section 11 of MIL-HDBK-217F Notice 2.

2.1.6.1 Ferrite Beads

Observation 2.1.6.1-1: Ferrite Beads are not included in MIL-HDBK-217F Notice 2.

Ferrite beads are made up from ferrite sheets with a printed electrode pattern, the sheets are stacked to form a chip inductor additional turns are achieved using via holes between sheets. By construction process they somewhat resemble a ceramic capacitor; however in the absence of a reliability model specific to a ferrite bead, use the inductor model in section 11 of MIL-HDBK-217F Notice 2. Please note that a chip inductor model is absent in proprietary prediction software, for this reason do not select chip in the inductor subcategory or the failure rate will be set to zero.

To reproduce the inductor values in the parts count table page A-9 Section 11.2 Coil, Fixed Inductor or Choke use the following in the formula $\lambda_p = \lambda_b * \pi T * \pi Q * \pi E$. Use the value of $\lambda_b = 0.00003$ fpmh, and formula for $\pi T = \text{EXP}((-0.11/(8.617*10^{-5})*(1/(T+273)-1/298)))$ where T is the environment temperature (neglect any temperature rise), $\pi Q = 1$ (the recommended value of 3 would be used subsequently) and πE is found in the table on page 11-3 in section 11.2 of MIL-HDBK-217F Notice 2.

2.1.7 Relays

Observation 2.1.7-1: Relays can be found in section 13 of MIL-HDBK-217F Notice 2.

Rule 2.1.7-1: “Quality Level” PiQ = 1.5 SHALL be used as a standard default for commercial relays, mechanical, of known pedigree. (Category 1)

Rule 2.1.7-2: “Quality Level” PiQ = 1 SHALL be used as a standard default for commercial relays, solid state and time delay, of known pedigree. (Category 1)

2.1.8 Switches

Observation 2.1.8-1: Switches can be found in section 14 of MIL-HDBK-217F Notice 2.

Rule 2.1.8-1: “Quality Level” PiQ = 1 SHALL be used for commercial switches of known pedigree. (Category 1)

2.1.9 Connector

Observation 2.1.9-1: Connectors can be found in section 15 of MIL-HDBK-217F Notice 2.

Rule 2.1.9-1: “Quality Level” PiQ = 1 SHALL be used as a standard default for commercial connectors, general, of known pedigree. (Category 1)

Rule 2.1.9-2: Connector “Pairing” = Single Connector SHALL be used if modeling a module or Circuit Card Assembly (CCA).

Rule 2.1.9-3: “Connector Type” = Rectangular SHALL be used as a standard default for commercial connectors with a subcategory of “General.”

Rule 2.1.9-4: “# of Mating Cycles” = 0 to 0.05 SHALL be used as a standard default for commercial connectors. (Category 1)

Permission 2.1.9-1: The failure rate MAY be adjusted based on functionality considerations. A connector that is only used for testing, for example, MAY be excluded from the system failure rate so long as care is taken to consider any failure modes that MAY affect the system, for example shorts.

2.1.10 Interconnection Assemblies, Surface Mount Technology

Observation 2.1.10-1: Interconnection Assemblies can be found in section 16 of MIL-HDBK-217F Notice 2

Rule 2.1.10-1: For the Lead Configuration Factor (PiLC), PiLC =100 SHALL be used for plastic BGAs and PiLC=50 SHALL be used for ceramic BGAs. (Category 1)

Rule 2.1.10-2: “Quality Level” PiQ = 1 SHALL be used as a standard default for commercial interconnection assemblies of known pedigree. (Category 1)

Observation 2.1.10-2: The current MIL-HDBK-217F Notice 2 model for calculating the failure rate of interconnection assemblies with plated-through holes (PTH), implies that the higher the quantity of PTH and/or circuit planes, the higher the interconnection assembly failure rate λ_p . This model results in an unrealistically high λ_p when dealing with large numbers of PTH and/or circuit planes. There are instances when applying this model produces an interconnection assembly failure rate that is an order of magnitude larger than all of the components mounted onto the printed wiring assembly (PWA), including the most complex microcircuit.

Recommendation 2.1.10-1: An alternative method of determining the failure rate of interconnection assemblies with plated-through holes is to use a physics-of-failure (PoF) analytical approach. Scientific testing and data indicates that PTH strain-driven fatigue failure, to the extent that it will occur in service, is the dominant interconnection assembly failure mechanism [7]. The PoF model for PTH fatigue failure is provided in Appendix F.

Permission 2.1.10-1: The number of PTH used in the MIL-HDBK-217F Notice 2 model MAY be limited to active pins. Inactive pins, or those covered by redundancy, MAY be excluded so long as they do not have circuit trace connections.

2.1.11 Meters

Observation 2.1.11-1: Meters can be found in section 18 of MIL-HDBK-217F Notice 2.

Rule 2.1.11-1: “Quality Level” PiQ = 1 SHALL be used as a standard default for commercial meters of known pedigree. (Category 1)

2.1.12 Crystal

Observation 2.1.12-1: Crystals can be found in section 19 of MIL-HDBK-217F Notice 2

Rule 2.1.12-1: “Quality Level” PiQ = 1 SHALL be used as a standard default for commercial crystals of known pedigree. (Category 1)

2.1.13 Oscillator

Observation 2.1.13-1: Oscillators can be found in section 19 of MIL-HDBK-217F Notice 2

Rule 2.1.13-1: Category = “Miscellaneous” and subcategory = “Quartz Crystal” SHALL be used as a standard default for oscillators. (Category 1)

Rule 2.1.13-2: “Quality Level” PiQ = 1 SHALL be used as a standard default for commercial oscillators of known pedigree. (Category 1)

2.1.14 Electronic Filters

Observation 2.1.14-1: Electronic Filters can be found in section 21 of MIL-HDBK-217F Notice 2.

Rule 2.1.14 -1: “Quality Level” $PiQ = 1$ SHALL be used as a standard default for commercial electronic filters of known pedigree. (Category 1)

2.1.15 MEMS

Observation 2.1.15-1: MEMS are not included in MIL-HDBK-217F Notice 2.

2.1.15.1 MEMS Oscillators

Observation 2.1.15.1-1: MEMS Oscillators are not included in MIL-HDBK-217F Notice 2. An example MEMS Oscillators calculation is provided in Appendix G.

2.1.15.2 MEMS Hybrids

Observation 2.1.15.2-1: MEMS Hybrids are not included in MIL-HDBK-217F Notice 2.

2.2 Parts Count Method Standard Defaults

For each standard default provided in this section, the categories defined in 1.3.3 are noted to clarify the quality of the derivation.

2.2.1 General

Rule 2.2.1-1: The reliability prediction SHALL be associated with a stated environment.

Observation 2.2.1-1: The specified environment will determine a default temperature, within the parts count method.

2.2.2 Microcircuits

Rule 2.2.2-1: Quality Factor of $PiQ = 1$ SHALL be used as a standard default for commercial Microcircuits of known pedigree. (Category 1)

Rule 2.2.2-2: Learning Factor > 2.0 years ($PiL = 1.0$) SHALL be used as a standard default for Microcircuits. (Category 1)

2.2.3 Discrete Semiconductors

Suggestion 2.2.3-1: Use the defaults in MIL-HDBK-217F Notice 2 Appendix A, for the part types and packaging being utilized.

Rule 2.2.3-1: Quality Level ($PiQ = 1$) SHALL be used as a standard default for commercial semiconductors of known pedigree. (Category 1)

2.3 Adjustment Guidelines

In the previous sections, standard default values were established for a majority of the part types used in electronic systems. The default values are intended to be a starting point in developing part models for reliability prediction purposes. If little or no part information is available, the default values should be used. Modifications to the default values can be implemented provided there is sufficient justification behind each part model modification. All changes to the default values will be documented, either within the details of the reliability prediction report or in the “Reliability Prediction Disclosure Statement” (reference VITA 51.0 Rule 2.5-1) that will accompany the report. The documented changes include the modified value used and the justification or basis for using it.

Since default value modifications are likely to occur, some direction is necessary to make sure that the changes made are reasonable and consistent as possible. The remainder of this section attempts to provide guidance when making adjustments to the default values.

This section applies to either parts count or parts stress analyses.

2.3.1 Module Design

A reliability engineer may not always have all the information relating to how a particular module was designed. Because of this, selecting the appropriate stress ratio, such as voltage or power ratio, may be difficult. A default value was specified for the related part types in an attempt to be average. If the reliability engineer has access to the detailed module design information and part application notes, then the various stress ratios should be modified to reflect the actual stress ratio. (see Recommendation 2-1) This would need to be performed on each part, as appropriate. For example, if a capacitor is rated at 20V and the applied operating voltage is 5V, then the voltage (stress) ratio is 0.25 (25%) versus the default value of 0.5 (50%).

2.3.2 Quality Factor

Generally speaking, MIL-HDBK-217F Notice 2 is overly pessimistic when assigning quality factors to commercial parts. Section 2 specifies lower quality factor values that can be used to better reflect the higher quality parts being manufactured today. The quality factor can vary from one commercial supplier to another for the same family of parts. The default values assume that one knows little about the supplier, but action is taken to make sure the parts are appropriate for the application. The quality factor can be lowered from the default value provided there is evidence to justify the change, such as knowing more about the supplier's quality program and the level of controls that are implemented.

2.3.3 Utilizing Field Experience

Use of field experience data, either to supplement a reliability prediction or to replace the prediction provides the potential of a more realistic reliability assessment than a handbook prediction.

However, field experience data analysis introduces new difficulties and limitations, that if not properly addressed can lead to very inaccurate results and misleading conclusions. The main difficulty in using field experience is collecting a complete and well characterized data sample from the field. If the user is depending solely on customer reporting, there may be lapses and inaccuracies in reporting that will be both unknown and uncontrollable. This problem may be overcome, by using embedded sensors or some type of automated reporting.

Field experience data may also be proprietary, either to the supplier company or for customers due to concerns about liability or public relations. Reliability analyses based on limited or company proprietary information may not be able to be substantiated by an independent analysis, and consequently may be less acceptable as a deliverable product. This may be overcome by establishing data sharing agreements.

The use of field data can provide a more realistic reliability assessment, and be more useful for maintenance planning, warranties and when providing a reliability prediction to a customer. For these reasons, it's important to document and communicate the use of field data to the customer. A reliability prediction report that uses field data must include a description of its use on the "Reliability Prediction Disclosure Statement" (VITA 51.0 Rule 2.5-1).

2.3.4 Manufacturer Test Data

Incorporating a manufacturer's data into a reliability prediction can be a good way of keeping up with the improvements in reliability and advances in technology. Some manufacturers have excellent reliability data resources which are quite easy to access. With some manufacturers, the data is available but more difficult to locate, whilst others will require a special request.

Rule 2.3.4-1: Manufacturer test data SHALL be adjusted for the usage temperature and environments and the method used SHALL be documented.

Permission 2.3.4-1: For the microcircuit failure rate prediction models of MIL-HDBK-217F Notice 2 section 5, the only Pi factors that are variable for a given device are those for the prediction temperature and prediction environment. The formula below provides a method of adjusting the manufacturer's established reliability numbers relationally to the temperature and environment adjustments between the manufacturers' testing and prediction conditions.

For digital logic manufacturer data, the following conversion equation MAY be used to correct for the usage temperature (π_T) and environments (π_E). Use the MIL-HDBK-217F Notice 2 π_T and π_E factors to adjust vendor data (λ_{BMFR}) from the specified temperature (π_{TMFR}) and ground benign (π_{EMFR}) test conditions to the usage operating environment:

$$\lambda_P = \left(1 + \frac{\pi_T - \pi_{TMFR}}{\pi_{TMFR}} + \frac{\pi_E - \pi_{EMFR}}{\pi_{EMFR}} \right) \lambda_{BMFR}$$

Permission 2.3.4-2: The conversion factors available in the RAC Reliability Toolkit: Commercial Practices Edition [8] MAY be used to correct for a different environment.

Observation 2.3.4-1: At the time of the writing of this specification, an updated copy of the RAC Reliability Toolkit could be obtained from the System Reliability Center (SRC) website.

Permission 2.3.4-3: The Parts Count failure rates in MIL-HDBK-217F Notice 2 Appendix A MAY be adjusted using manufacturer data with their published environment and temperature factors. The Parts Count values were produced by applying default values to the equations in MIL-HDBK-217 Section 5. Because of this, where C1 and C2 remain the same for that default, only PiT and PiE change according to the temperature and environment; C1 and C2 are built into the equations. Therefore, to convert manufacturer's data, typically provided at 55 degrees C (and resulting from tests conducted without vibration), to any other environment, it is only necessary to find the ratio between the manufacturer's failure rate and the Ground Benign, 30C ambient temperature (Ground Benign, 50C junction temperature) value from the row most appropriate to the device. An example of this is provided in this document's Appendix H.

2.3.5 Mixing Models and Methods

Recommendation 2.3.5-1: Mixing models, i.e. using different reliability prediction methodologies (e.g. 217Plus, Telcordia, etc.) for some components, SHOULD NOT be done for any technology or component that has a model in MIL-HDBK-217F Notice 2.

Permission 2.3.5-1: If no model exists in MIL-HDBK-217F Notice 2 for a technology or component, a different modeling method may be used for that technology or component.

Rule 2.3.5-1: Mixing methods, i.e., combining wearout and random failure rate models shall only be done in accordance with the ANSI/VITA 51.2 section on mixing methods.

Observation 2.3.5-1: Improper combining of wearout and random failure rate metrics may lead to an inaccurate MTBF estimate. Another problem arises if wearout is used in place of MTBF or used as the only reliability prediction metric. A discussion of why this is so is in Appendix I.

3 Appendix A: Acknowledgements

3.1 Working Group Members – Original Specification

The following individuals have participated in the creation of this specification and are gratefully acknowledged:

Participants:

Name	Affiliation
Lori Bechtold (Chair, Editor)	Boeing
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Dan Quearry (Vice Chair)	Naval Surface Warfare Center, Crane
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Table 3-1: Working Group Members – Original Specification

3.2 Working Group Members – Revision 1

The following individuals have participated in the creation of this specification and are gratefully acknowledged:

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Table 3-2: Working Group Members – Revision 1

3.3 ANSI Ballot Group

The following individuals have participated in the balloting of this specification and are gratefully acknowledged:

Name	Affiliation
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Barrett Bonner	Lockheed Martin Space Systems Company
Gene Bridgers	Mercury Systems
Daniel Brown	Lockheed Martin Aeronautics

Chris Eckert	GE Aviation Systems
Thomas Faltus	Boeing
Travis Fullem	Bechtel Marine Propulsion Corporation
Lou Gullo	Raytheon
Heiko Koerte	N.A.T. GmbH
Peter Miskelly	GE
Mark Mooder	Curtiss-Wright Controls
Robert Morley	DRS Technologies
Murray Nash	General Dynamics Canada Ltd.
Scott Newland	Harris
Marcus Niessen	Lockheed Martin
Danny Quearry	SDA/NSWC Crane
John Ryneerson	VITA
Michael Shorey	Mercury Systems
Hermann Strass	Technology Consulting
Bahig Tawfellous	Honeywell Aerospace
Michael Thompson	Pentair/Schroff
Kevin Thorson	Lockheed Martin Corporation

Table 3-3: ANSI Ballot Group

4 Appendix B: Counterfeit Parts

Counterfeit parts have been an increasing serious problem over the last few years. Counterfeit parts have an unknown failure rate that will cause an inaccurate and inconsistent reliability prediction. This appendix contains information that can help the reliability engineer understand some of the problems and includes a list of organizations that are working the counterfeit parts issues. The system that the prediction is being performed on should have a counterfeit parts control plan to detect and remove counterfeit parts.

The purpose of this document is to establish and maintain consistent and uniform methods for estimating the inherent reliability (i.e., the reliability of a mature design) of high reliability and military electronic equipment and systems. It provides a common basis for reliability predictions during acquisition programs for military electronic systems and equipment. It also establishes a common basis for comparing and evaluating reliability predictions of related or competitive designs. The handbook is intended to be used as a tool to assess the reliability of the equipment being designed. This document does not address performing reliability predictions with the use of parts that are counterfeit or from unknown sources.

On May 21, 2012, the Senate Armed Services Committee (SASC) released the final report of its "Inquiry Into Counterfeit Electronic Parts in the Department of Defense Supply Chain." The report followed a two-year investigation and draws upon hearings conducted by the SASC in 2011. Those hearings, in turn, led to Section 818 of the FY 2012 National Defense Authorization Act, a new law intended to stop importation of counterfeit parts, require the DOD and contractors to improve purchasing practices to favor original manufacturers and "trusted suppliers," and compel regular and diligent reporting when counterfeit parts are identified or suspected. Section 818 also makes contractors responsible for the costs of rework or corrective action as required to redress counterfeit and suspect counterfeit parts.

Organizations that are working Counterfeit parts:

SAE G-19 committee is heavily involved with counterfeit parts.

AIA is also, to a much lesser extent.

University of Maryland CALCE

Specifications that cover Counterfeit Parts:

SAE Aerospace Standard AS5553, Counterfeit Electronic Parts; Avoidance, Detection, Mitigation, and Disposition was developed by the G-19 committee of SAE International.

SAE AS6081, Fraudulent/Counterfeit Electronic Parts: Avoidance, Detection, Mitigation, and Disposition – Distributors.

SAE AS9100 Quality Management System Requirements – Aerospace

SAE AS6174 (non-electronic parts)

SAE ARP6178 (supplier assessment)

SAE AS6171 is being developed but it won't be out for another year.

MIL-STD-3018 Parts Management

IEC 62668-1 (Process management for avionics – Counterfeit prevention)

IEC 62239 (Electronic Components Management Plan)

IEC 62240 (Use of semiconductor devices outside manufacturers' specified temperature ranges)

5 Appendix C: Guidelines for “Known Pedigree”

Known Pedigree components are defined as meeting some minimum criteria. This will not necessarily mean best practices. For the purpose of the use of factors within this specification these are the minimum acceptable criteria for defining a component as having a known pedigree.

- 1) Components that are ECMP [1] or AQEC [2] qualified are considered to be known pedigree.
- 2) Parts that are procured in accordance with a user parts management program which is approved by their customer are considered to be known pedigree.
- 3) For all other components, answer the following questions:
 - a) Is the supplier ISO-9000 certified?
 - b) Has this supplier been evaluated (surveyed, visited) and qualified by the user/contractor?
 - c) Does the supplier publish reliability data periodically?
 - d) Does the supplier have controls in place to manage their materials?
 - e) Does the supplier provide timely notification of changes of material or processes?
 - f) Does the supplier have a Failure Analysis or FRACAS system?
 - g) Does the supplier have a configuration management system?
 - h) Does the supplier have a quality management system?

If the answer to all the above questions is “yes” then the component is considered to be known pedigree.

If a parts supplier doesn’t meet any of the above criteria, then it is the responsibility of the user of this specification to justify the use of quality factors provided for the known pedigree parts.

References:

- [1] IEC-TS-62239, Standard for Preparing an Electronic Components Management Plan
- [2] GEIA-STD-0002-1, The Aerospace Qualified Electronic Components (AQEC) Requirements

6 Appendix D: MOSFETs

The base failure rate for MOSFET in MIL-HDBK-217F Notice 2 section 6.4 (Low frequency < 400 MHz) is 0.012×10^{-6} and section 6.9 (High frequency > 400 MHz) is 0.060×10^{-6} . Each was derived more than 20 years ago; therefore an update to the base failure rate was performed.

The approach to update the base failure rate was by pursuing at least one reputable MOSFET vendor and obtaining their reliability test data. All reputable vendors perform reliability testing today; the challenge is to find vendors who are willing to share the data.

Transistor manufacturers typically perform their high temperature testing at 85°C to 150°C. Once the test data is obtained, the data needs to be normalized to 25°C since it is the temperature used by MIL-HDBK-217F Notice 2 to normalize all base failure rates.

25°C is where all base failure rates were normalized to in MIL-HDBK-217F Notice 2. The method used to normalize the current vendor data was using the software tool by Texas Instruments {search for the tool ("Reliability Data: Reliability Estimator Results") at <http://www.ti.com/>}, which allows conversions to occur from just about any temperature and from any confidence level or activation energy desired.

For the MOSFET base failure rate calculation, 0.7 eV was used for the activation energy, and a 60% confidence level was used, which conforms to how MIL-HDBK-217F Notice 2 factors were calculated.

The aggregated data, for both low and high frequency MOSFETs, resulted in a base failure rate of 0.0006 FPMH which is two orders of magnitude less than the base failure rate currently listed in MIL-HDBK-217F, Notice 2, for high frequency MOSFETs. The VITA51 working group decided to be conservative and use the data to justify reducing the base failure rate by only one order of magnitude to 0.0060 FPMH for the high frequency MOSFETs (Recommendation 2.1.3-2). Similarly, the base failure rate for low frequency MOSFETs is reduced to 0.0012 FPMH (Recommendation 2.1.3-1).

Supporting vendor data and calculations can be found in this [spreadsheet](#).

The spreadsheet may also be used to try different "what if scenarios" for various inputs such as usage temperature, different confidence levels, or FIT rates.

7 Appendix E: Tantalum Capacitors

A long view of Tantalum Capacitors

Vishay Sprague is well known for having invented the first tantalum capacitors and the first surface-mount tantalum chip capacitors.¹ Tantalum capacitors were first used in the 1950's and emulated the construction of aluminium electrolytic with foil construction. Sintered construction enabled the manufacture of surface mount tantalum chip capacitors.²

However although tantalum capacitors are available down to 1nF, values below 100nF are axial leaded while surface mount capacitors are only available from 100nF.³ Since this model is for CWR equivalent surface mount devices, the smaller values may be excluded from consideration.

MIL-HDBK-217F Notice 2 was released in 1995.

In the QPL, 20th August 1998, CWR06 – 09 no values given, then CWR11 begins at the lowest value of 0.1uF.⁴

Applying this knowledge to the data

Looking at the assumptions made for the reliability prediction using field data, initially the starting value used for capacitance was 0.0022 uF or 2.2 nF, the default used to create the failure rates given in Appendix A of MIL-HDBK-217F Notice 2.⁵ This resulted in a particularly low failure rate caused by the value of PiC for this capacitance. By examining how PiC varies with capacitance, it can be seen that low values of C give rise to particularly low values of PiC.^{6, 7}

When this predicted MTBF is compared to the calculated MTBF it gives the wrong impression, that the calculated MTBF is worse than the predicted MTBF. However it is not possible to buy SMT capacitors in such small values today, if it ever was. The smallest available capacity in a tantalum capacitor is 0.1 uF or 100 nF. The value used by 217 as a default is 0.0022 uF or 2.2 nF, 45 times smaller than 100 nF.

If 100 nF (the smallest available value) is assumed for missing values in the data in the prediction, then the field values are better than the predicted values and the prediction is cautious (should 10uF be used it is more so). Then using 50% derating as a default a PiQ of 0.064 will match the predicted value to the field value. This suggests that 0.1 as a PiQ is not unreasonable.

Carrying out the prediction on the CWR data

In the associated Excel spreadsheet VITA_TANT_217F_N2_PARTS STRESS_MAR_2013.xlsx, the complete field data set is available in the worksheet VITA CAPACITORS ALL. From this the two sets of data relating to tantalum chip capacitors were combined and listed in the worksheet TANTS ONLY1. In this worksheet a simple reliability calculation was performed to yield a field data MTBF of 7,749,806 hours.

The formula used was $\lambda_D = \lambda_b * \pi_T * \pi_C * \pi_V * \pi_{SR} * \pi_Q * \pi_E$ from page 10-1 in Section 10.1 Capacitors in MIL-HDBK-217F Notice 2. The capacitor style is CWR to MIL-C-55365.

Lambda b (failure rate given) is 0.00005 FPMH,

PiT uses column 1 where Ea is 0.15 V and the formula in Excel is =EXP((-0.15/(8.617*10^-5))*(1/(T+273)-1/298)) where T is the ambient temperature. The values for T used in the calculation were the default temperatures for the environments listed AIC, ARW and AUF are 55, 55 and 70 (°C) respectively, (assumption 1).

PiC is given by =C^0.23 where C is the value in uF and an exploration is outlined in the Appendix. For those components without recorded capacitance values, an assumed value of 0.1 uF was used, (assumption 2).

PiV uses column 4 and is given by =(0.5/0.6)^17+1 where 0.5 is the assumed stress ratio, (assumption 3).

PiSR is assumed to be 1, (assumption 4).

PiQ is the variable and is changed to match the predicted failure rate with the calculated field failure rate.

PiE is appropriate to the environment, for AIC, ARW and AUF these are 12, 40 and 30 respectively from the table on page 10-5 in Section 10.1 in MIL-HDBK-217F Notice 2.

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The product for each entry is also multiplied by the given quantity, and then the resulting failure rates are summed, assuming serial reliability.

The total on the page is 0.201605 FPMH, equivalent to 4,960,205 hours MTBF.

This may be compared with the field failure rate of 0.129035485 FPMH equivalent to 7,749,806 hours MTBF.

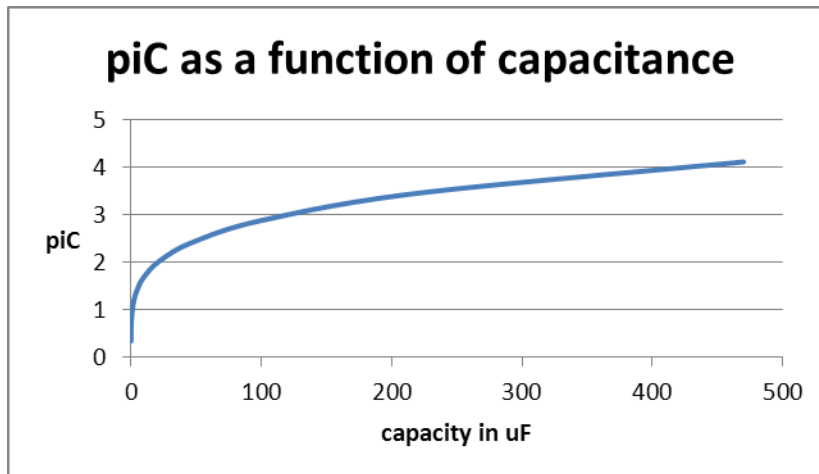
Within the worksheet captant_CWR MATCH IT, by iteration of the value in cell H45 it turns out that a PiQ of 0.064 will match the predicted value to the field value.

For exploration purposes the value (in uF) in cell J46 may be changed and will update all the unknown entries.

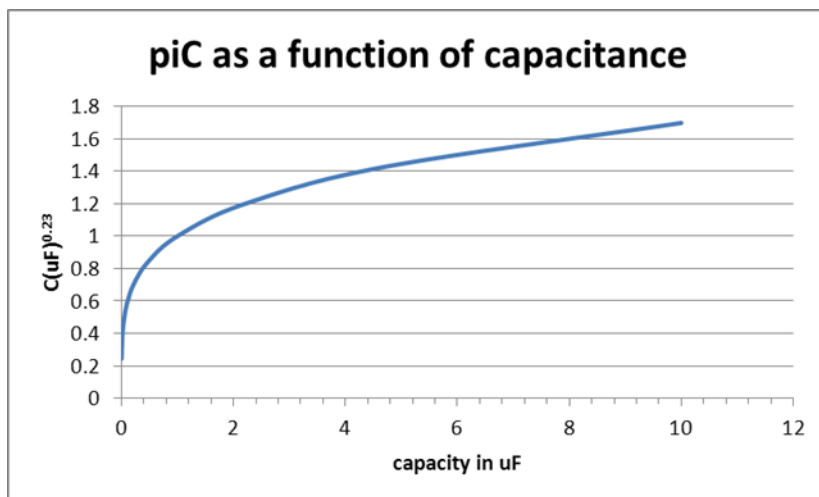
FAQ⁸

REFERENCES

- 1) <http://www.vishay.com/company/brands/sprague/>
- 2) <http://www.avx.com/docs/techinfo/bsctant.pdf>
- 3) <http://www.vishay.com/capacitors/tantalum/capacitance-1nf-to-100nf/>
- 4) <http://www.dsccl.dla.mil/downloads/qplqml/55365/archive/55365rev057.pdf>
- 5) Value derived from Relex.



6)



7)

8) Any questions may be addressed to peter.miskelly@ge.com

9) 8th March 2013

8 Appendix F: A Physics-of-Failure Prediction Approach for Interconnection Assemblies with Plated-Through Holes

The current *Military Handbook (MIL-HDBK)-217 F, Notice 2* model for calculating the failure rate of interconnection assemblies (IA) with plated-through holes (PTH) [1], shown as Equation 1.1 below, implies that the higher the quantity of PTH and/or circuit planes, the higher the IA failure rate λ_p . This model results in an unrealistically high λ_p when dealing with large numbers of PTH and/or circuit planes. There are instances when applying this model produces an IA failure rate that is an order of magnitude larger than all of the components mounted onto the printed wiring assembly (PWA), including the most complex microcircuit.

$$\lambda_p = \lambda_b [N_1 \pi_c + N_2 (\pi_c + 13)] \pi_Q \pi_E \text{ Failures}/10^6 \text{ Hours} \quad \text{Equation 1.1}$$

Where

- λ_b = Base Failure Rate [$\lambda_b = 0.000017$ for IA with PTH and $\lambda_b = 0.00011$ for discrete wiring with electroless deposited PTH (≤ 2 levels of circuitry)]
- N_1 = Quantity of Wave-Soldered PTH
- N_2 = Quantity of Hand-Soldered PTH
- π_c = Complex Factor Based on the Number of Circuit Planes P [look-up chart; e.g., $\pi_c = 4$ for $P = 18$]
- π_Q = Quality Factor [$\pi_Q = 1$ for MIL-SPEC or Packaging Electronic Circuits (IPC) Standards IPC Level 3, and $\pi_Q = 2$ for lower quality]
- π_E = Environment Factor (look-up chart; e.g., $\pi_E = 28$ for Airborne Uninhabited Fighter (AUF))

An alternative method of determining the failure rate of interconnection assemblies with plated-through holes is to use a physics-of-failure (PoF) analytical approach. Scientific testing and data indicates that PTH strain-driven fatigue failure, to the extent that it will occur in service, is the dominant IA failure mechanism [2]. The PoF model for PTH fatigue failure of IA consists of three steps:

- **STEP 1:** Solve for σ , the load, using the following formula [2]

$$\sigma = \frac{[(\alpha_1 - \alpha_2) \Delta T + S_y \frac{E_2 - E'_2}{E_2 E'_2}]}{A_1 E_1 + A_2 E'_2} \quad \text{Equation 1.2}$$

Where

- α_1 is the coefficient of thermal expansion (CTE) of the IA material in the z direction (NOTE: Values for various materials are provided in Table 1)
- α_2 is the coefficient of thermal expansion of the PTH barrel material, Cu, $\sim 1.8 \times 10^{-5}$ in/in/°C
- ΔT is the operating temperature range (°C)
- S_y is the yield strength of PTH barrel material, Cu, $\sim 2.5 \times 10^4$ psi

- A_1 is the IA area influencing the deformation of the PTH barrel [2]

$$A_1 \cong \frac{\pi}{4} [(h + d)^2 - d^2] \quad \text{Equation 1.3}$$

- ❖ d = drilled PTH diameter (inches)
- ❖ h = thickness of IA (inches)

****NOTE:** d is best determined by thermal analysis, where the PTH with the smallest diameter within the hottest spot on the PWA is chosen. If no thermal analysis data is available, use the PTH with the smallest diameter on the board.

- A_2 is the area of the PTH barrel [2]

$$A_2 \cong \frac{\pi}{4} [d^2 - (d - 2t)^2] \quad \text{Equation 1.4}$$

- ❖ d = drilled PTH diameter (inches)
- ❖ t = thickness of PTH diameter (inches)

****NOTE:** which t to select is best determined through thermal analysis. Select the PTH with the smallest diameter within the hottest spot on the PWA. If no thermal analysis data is available, use the PTH with the smallest diameter on the board.

- E_1 is the elastic modulus of the IA material (**NOTE:** Values for various materials are provided in Table 1)
- E_2 is the elastic modulus of the PTH barrel material, Cu, $\sim 1.2 \times 10^7$ psi
- E'_2 is the plastic modulus of the PTH barrel material, Cu, $\sim 0.1 \times 10^6$ psi
- **STEP 2:** Solve for $\Delta \epsilon$, the total cyclic-strain range, using Equation 1.5 [2]

$$\Delta \epsilon = \frac{S_y}{E_2} + \frac{\sigma - S_y}{E'_2} \quad \text{Equation 1.5}$$

Where

- S_y is the yield strength of PTH barrel material, Cu, $\sim 2.5 \times 10^4$ psi
- σ is the load determined from STEP 1
- E_2 is the elastic modulus of the PTH barrel material, Cu, $\sim 1.2 \times 10^7$ psi
- E'_2 is the plastic modulus of the PTH barrel material, Cu, $\sim 0.1 \times 10^6$ psi
- **STEP 3:** The expected number of cycles-to-failure $\bar{N}_f$ can then be predicted by solving the following equation [3] iteratively or by using a math software tool.

$$\Delta \epsilon = \bar{N}_f^{-0.6} D_f^{0.75} + 0.9 \frac{S_u}{E_2} \left[\frac{\exp(D_f)}{0.36} \right]^{0.1785 \log \frac{10^5}{\bar{N}_f}} \quad \text{Equation 1.6}$$

Where

- $\Delta \epsilon$ is the total cyclic-strain range found in STEP 2
- $\bar{N}_f$ is the expected number of cycles-to-failure (fatigue life)

- D_f is the ductility constant of the PTH barrel material, Cu, ~30%
- S_u is the ultimate strength of the PTH barrel material, Cu, ~4 x 10⁴ psi
- E_2 is the elastic modulus of the PTH barrel material, Cu, ~1.2 x 10⁷ psi

Based on the number of hours per thermal cycle (H_c) of the system in which the IA is utilized, the failure rate per million hours (**FPMH**) can then be determined by the following formula:

$$FPMH = \frac{10^6}{H_c \times N_f} \quad \text{Equation 1.7}$$

Material	Elastic Modulus E_1 (psi, x10 ⁶)	CTE in z-direction α_1 (in/in/°C, x10 ⁻⁵)
Epoxy Aramid	3.19 - 4.06	10.5
Epoxy Glass (FR-4, G-10)	2.52	6.0 - 8.0
Epoxy Quartz	2.7	6.2
Polyimide Aramid	2.9 - 3.92	8.3
Polyimide Glass	2.84	6.0 - 8.0
Polyimide Quartz	4.0	3.5
PTFE Glass	0.145	26.1

Table 1 Properties of Selected IA Laminates [4,5]

REFERENCES

- [1] *Military Handbook (MIL-HDBK)-217 F Notice 2*, "Reliability Prediction of Electronic Equipment" February 28, 1995.
- [2] *IPC-TR-579*, "Round Robin Reliability Evaluation of Small Diameter Plated Through Holes in Printed Wiring Boards," The Institute for Interconnecting and Packaging Electronic Circuits, September 1988.
- [3] Engelmaier, W., and T. Kessler. "Investigation of Agitation Effects on Electroplated Copper in Multilayer Board Plated-Through-Hole in a Forced-Flow Plating Cell." *J. Electrochemical Soc.*, vol. 125, no. 1, January 1978.
- [4] Pecht, M., ed. *Handbook of Electronic Package Design*. New York: Marcel Dekker, Inc., 1991.
- [5] Pecht, M., A. Rakesh, P. McCluskey, T. Dishongh, S. Javadpour, and R. Mahajan. *Electronic Packaging: Materials and Their Properties*. New York: CRC Press LLC, 1999.

9 Appendix G: MEMS Oscillators Calculations

Using the 5.1 Section of MIL-HDBK-217F Notice 2 and considering the device as a microcircuit and analog, since the device is crafted in silicon (only) and contains analog circuit elements.

Due to its construction, size and pin count, select the row of values from the fifth block, MOS Technology, Linear Microcircuits (Ea = .65) 1 to 100 Transistors (14 Pin DIP) 0.0095, 0.024, 0.039, 0.034, 0.049, 0.057, 0.062, 0.12, 0.13, 0.076, 0.0095, 0.044, 0.096.

In order to reproduce the values given on page A-2 in APPENDIX A: PARTS COUNT of MIL-HDBK-217F use the formula $(C1 \cdot PiT) + (C2 \cdot PiE) \cdot PiQ \cdot PiL$ found on page 5-3 in section 5.1 in the main body of MIL-HDBK-217F.

For C1 use 0.01 and for C2 use 0.0048415 (this is the result of using equation number 1 in Section 5.9 on page 5-14 $c2 = 2.8 \cdot 10^{-4} (Np)^{1.08}$ with $Np=14$

For PiT use the formula $= 0.1 \cdot \exp\left(\frac{(-0.65)}{(8.617 \cdot 10^{-5})} \cdot \left(\frac{1}{(T+273)} - \frac{1}{298}\right)\right)$ where T is the ambient temperature plus 30 degrees C and PiE is from a lookup table 5.10 / 5-15 FOR INTEGRATED CIRCUITS a PiQ and PiL of 1 will match the figures above.

Environment	PiE
GB	0.05
GF	2
GM	4
NS	4
NU	6
AIC	4
AIF	5
AUC	5
AUF	8
ARW	8
SF	0.5
MF	5
ML	12
CL	220

Table 9-1: Environmental Factors

To convert manufacturer's data across environments, assuming it is produced equivalent to GB environment, divide the GB50 value by the data to give a factor, then divide each environment by the factor to convert the data, since PiT is already built in to each calculation the factor is constant.

10 Appendix H: Conversion Methods for Manufacturer's Data

The figures in 217 Appendix A are produced by applying default values to the equations in 217 Section 5. Because of this, where C1 and C2 remain the same for that default, only PiT and PiE change according to the temperature and environment; C1 and C2 are built in to the equations. Therefore to convert manufacturer's data, typically provided at 55 degrees C (and resulting from tests conducted without vibration), to any other environment, it is only necessary to find the ratio between the manufacturer's failure rate and the Ground Benign, 30C ambient temperature (Ground Benign, 50C junction temperature) 217 value from the row most appropriate to the device.

For example: Manufacturer's data for a MEMS oscillator is given as 2 FIT at 55 degrees C with 65% confidence, this is 0.002 FPMH and considered to be equivalent to the GB50 value (for a GB30 environment). <http://www.sitime.com/products/datasheets/sit8103/SiT8103-datasheet.pdf>

For a parts count prediction, the closest match from Appendix A is to choose the row beginning .0095, .024 etc. for a MOS Linear 14 pin (hermetic) DIP.

To convert this failure rate from GB50 to AUF90 (for example) either use the formula:

$$=((C1*PiT_2)+(C2*PiE_2))/((C1*PiT_1)+(C2*PiE_1))$$

Or within the limits of a parts count prediction using Appendix A values, divide the 217 GB50 value 0.0095 by the manufacturer's failure rate 0.002 to give 4.75 then divide the AUF90 value of 0.13 by the factor 4.75 to give 0.027368421, round to 0.027 or use the full number. This factor is the same for converting from GB50 to any other environment; it is built-in to each failure rate across the standard environments.

Please note that within Excel that such rounding will not take place and the additional digits will yield 0.027691175. It is probably easier to retain the full values and only round at the end.

To reproduce the GB50 value of 0.0095, it may be calculated using the formula $(C1*PiT)+(C2*PiE)$.

Where C1 = 0.01 (1 to 100 transistors), C2 = 0.0048415

C1 is from a lookup table in Section 5.1 on page 5-3 of 217.

C2 is calculated from the formula in Table 5.9 on page 5-14, using formula 1 where

$C2 = 2.8 * 0.0001 * (14^{1.08}) = 0.0048415$ this is abbreviated to 0.0048 in the first column of Table 5.9.

Note that in the above equation, 14 is the parts count default value for the number of (active) pins.

PiT is given by $= 0.1 * \text{EXP}(((0.65)/(8.617 * 10^{-5})) * (1/(50+273)-1/298)) = 0.7093641$

Note that in the above equation the value 0.65 is the Effective Activation Energy, Ea in electron Volts used for linear microcircuits (column 6 from Table 5.8 on page 5-13) and 50 is the junction temperature Tj for an ambient environment of 30 degrees C.

So $(0.0048415 * 0.7093641) + (0.0048415 * 0.5) = 0.009514371$

Note that in the above equation 0.5 is the PiE for GB from Table 5.10 on page 5-15 of 217.

Another example, a complex MOS logic device with a pin count greater than 224. The manufacturer's data is supplied under an NDA as 0.027 FPMH. No extrapolation is applied and for a parts count prediction, the closest match from Appendix A is to choose the sixth row from the fourth block, MOS Technology, Gate / Logic Arrays, Digital (Ea = 0.35) 30,000 to 60,000 Gates beginning .13, .31 etc.

The ratio of 217 to MD is $0.131633313 / 0.027 = 4.87531$ or using the truncated values in 217 it is $0.13 / 0.027 = 4.815$. To convert this to any default environment such as GM65 take the value for this row, 0.53 divide by the factor 4.815 and the manufacturer's converted data is 0.11 FPMH at GM65 (equivalent to a GM45 environment).

Other acceptable methods of environment conversion are to use the conversion factors within the Reliability Toolkit: Commercial Practices Edition. This has translation factors for "estimating" the relative impact of an item of equipment that has field or predicted data at one set of conditions and needs to be estimated at another set of conditions. The translation factors in the toolkit account for three different

types of situations: quality change, temperature change, and environment change. Quality and temperature translations are single parameter considerations. The environmental translation accounts for more than one parameter. Specifically, it accounts for temperature change, vibration increase or decrease and humidity variation. Therefore, given that the translation applied is an environmental change, this means it already has the temperature impact and reapplying the temperature will result in a far more pessimistic translation. Since the tool is only an approximation, it is recommended that one should follow this approximation with real test data or by performing a detailed stress reliability analysis at the conditions specified. One should review these data to see if all components actually meet the operating and non-operating envelopes.

11 Appendix I: Wearout and Random Failure

MIL-HDBK-217 F Notice 2 primarily calculates failure rates or mean time between failures (MTBFs) that are presumed to be in the useful life period, also known as the flat part of the bathtub curve. There are exceptions, like the models in section 16.2. However, for the majority of models in MIL-HDBK-217 F Notice 2, the failure rate is assumed constant and not changing over time. To support this assumption, the design activity must assure two things: 1) that there is a good program to control manufacturing defects (e.g. environmental stress screening) to limit infant mortality failures, and 2) that the expected life of the product in the usage environments is characterized and managed (i.e., understand and control wearout). ANSI/VITA 51.3 provides guidelines and standard practices to assure the validity of the constant failure rate assumption.

MTBF is often confused with wearout metrics, such as expected life or time-to-failure. This leads to problems because wearout failure rates are time dependent, generally increasing over time in an aging population, while MTBF is constant over time. Wearout is also a metric tied to an individual unit, while MTBF is a population metric.

To help understand this, consider the analogy of a human population. People on average live for a certain period of time, say for example 80 years. At about 80 years, a person can be expected to have a higher probability of dying than earlier in life, and the dominant failure mechanisms are related to aging are wearout of key subsystems (circulation, breathing, cognitive, etc.). This is true of any individual person, regardless of the population that person is a part of. MTBF is meaningless for that individual, because his or her life expectancy is determined by acceleration or deceleration of wearout factors (e.g. eating right, getting enough exercise, etc.).

MTBF has meaning for working with populations. For example, someone may want to know how often people in an office building will call in sick or die. Understanding the rate of occurrence of these events within the population may be used to predict medical costs or costs of absences in a workplace. A workplace population is selected to avoid both the very young and very old, so should be characterized as a population in the useful life period. The causes of workplace illnesses are typically not related to aging but to random events such as contagious illness or accident. Year after year, the rate of occurrence of absences will stay the same so long as turnover keeps the population uniformly distributed with respect to age and there are no fundamental environmental changes such as the building developing mold and causing an overall increase in the number of illnesses. MTBF is useful in this case for spares planning (substitute workers), for maintenance cost (medical cost), cost of failure (life insurance costs) or for schedule interruption analysis (work disruptions).

An individual's life expectancy isn't affected by the population he or she is part of. Aging happens, regardless of other events in a person's life. Similarly, wearout occurs in electronics due to fatigue mechanisms. Random events, however, may occur at any time and affect young and old.

Problems arise when wearout and MTBF are confused, or when an event is mischaracterized when calculating metrics from empirical data.

There are also considerations when analyzing assemblies whose components may have varying ages or come from various populations. Assembly age has to be determined as a worst case of the oldest component. The component most susceptible to wearout will drive the replacement planning.

In some cases obsolescence will dominate more than wearout.